UNDERACTUATED ROBOTICS

Algorithms for Walking, Running, Swimming, Flying, and Manipulation

Russ Tedrake

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Note: These are working notes used for a course being taught at MIT. They will be updated throughout the Spring 2024 semester. Lecture videos are available on YouTube.

Previous Chapter | Table of contents | Next Chapter

CHAPTER 6
Model Systems with Stochasticity

My goals for this chapter are to build intuition for the beautiful and rich behavior of nonlinear dynamical system that are subjected to random (noise/disturbance) inputs. So far we have focused primarily on systems described by

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)) \quad \text{or} \quad \mathbf{x}[n+1] = f(\mathbf{x}[n], \mathbf{u}[n]).$$

In this chapter, I would like to broaden the scope to think about

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{w}(t)) \quad \text{or} \quad \mathbf{x}[n+1] = f(\mathbf{x}[n], \mathbf{u}[n], \mathbf{w}[n]),$$

where this additional input $\mathbf{w}$ is the (vector) output of some random process. In other words, we can begin thinking about stochastic systems by simply understanding the dynamics of our existing ODEs subjected to an additional random input.

This form is extremely general as written. $\mathbf{w}(t)$ can represent time-varying random disturbances (e.g. gusts of wind), or even constant model errors/uncertainty. One thing that we are not adding, yet, is measurement uncertainty. There is a great deal of work on observability and state estimation that study the question of how you can infer the true state of the system given noise sensor readings. For this chapter we are assuming perfect measurements of the full state, and are focused instead on the way that "process noise" shapes the long-term dynamics of the system.

I will also stick primarily to discrete time dynamics for this chapter, simply because it is easier to think about the output of a discrete-time random process, $\mathbf{w}[n]$, than a $\mathbf{w}(t)$. But you should know that all of the ideas work in continuous time, too. Also, most of our examples will take the form of *additive noise*:

$$\mathbf{x}[n+1] = f(\mathbf{x}[n], \mathbf{u}[n]) + \mathbf{w}[n],$$

which is a particular useful and common specialization of our general form. And this form doesn't give up much -- the disturbance on step k can pass through the nonlinear function f on step $k+1$ giving rich results -- but is often much easier to work with.

When implementing stochastic systems in Drake, we aim to be rigorous about

randomness, to facilitate advanced algorithms that are written against stochastic systems and to enable, e.g. random simulation roll outs but still enable deterministic replay (simulation is deterministic given the random seed).

6.1 The Master Equation

Let's start by looking at some simple examples.

Example 6.1 (A Bistable System + Noise)

Let's consider (a time-reversed version of) one of my favorite one-dimensional systems:

$$\dot{x} = x - x^3.$$

This deterministic system has stable fixed points at $x^* = \{-1, 1\}$ and an unstable fixed point at $x^* = 0$.

A reasonable discrete-time approximation of these dynamics, now with additive noise, is

$$x[n+1] = x[n] + h(x[n] - x[n]^3 + w[n]).$$

When $w[n] = 0$, this system has the same fixed points and stability properties as the continuous time system. But let's examine the system when $w[n]$ is instead the result of a *zero-mean Gaussian white noise process*, defined by:

$$\forall n, E\left[w[n]\right] = 0,$$
$$E\left[w[i]w[j]\right] = \begin{cases} \sigma^2, & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases}$$

Here σ is the standard deviation of the Gaussian.

When you simulate this system for small values of σ, you will see trajectories move roughly towards one of the two fixed points (for the deterministic system), but every step is modified by the noise. In fact, even if the trajectory were to arrive exactly at what was once a fixed point, it is almost surely going to move again on the very next step. In fact, if we plot many runs of the simulation from different initial conditions all on the same plot, we will see something like the figure below.

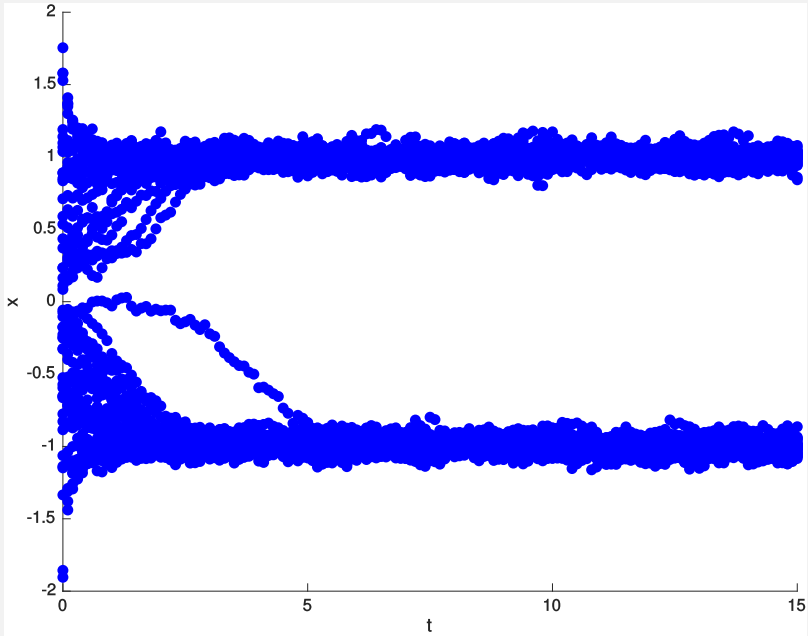

Figure 6.1 - Simulation of the bistable system with noise ($h = 0.1, \sigma = 0.3$) from many initial conditions.

During any individual simulation, the state jumps around randomly for all time, and could even transition from the vicinity of one fixed point to the other fixed point. Another way to visualize this output is to animate a histogram of the particles over time.

Figure 6.2 - Histogram of the states of the bistable system with noise ($\sigma = 1$) after simulating from random initial conditions until $t = 20$.

You can run this demo for yourself:

Let's take a moment to appreciate the implications of what we've just observed. Every time that we've analyzed a system to date, we've asked questions like "given x[0], what is the long-term behavior of the system, $\lim_{n\to\infty} x[n]$?", but now $x[n]$ is a *random variable*. The trajectories of this system do not converge, and the system does not exhibit any form of stability that we've introduced so far.

All is not lost. If you watch the animation closely, you might notice the *distribution* of this random variable is actually very well behaved. This is the key idea for this chapter.

Let us use $p_n(x)$ to denote the probability density function over the random variable x at time n. Note that this density function must satisfy

$$\int_{-\infty}^{\infty} p_n(x)dx = 1.$$

It is actually possible to write the *dynamics of the probability density* with the simple relation

$$p_{n+1}(x) = \int_{-\infty}^{\infty} p(x|x')p_n(x')dx',$$

where $p(x|x')$ encodes the stochastic dynamics as a conditional distribution of the next state (here x) as a function of the current state (here x'). Dynamical systems that can be encoded in this way are known as *continuous-state Markov Processes*, and the governing equation above is often referred to as the "master equation" for the stochastic process. In fact this update is even linear(!) ; a fact that can

enable closed-form solutions to some impressive long-term statistics, like mean time to failure or first passage times[1]. Unfortunately, it is often difficult to perform the integral in closed form, so we must often resort to discretization or numerical analysis.

Example 6.2 (The Logistic Map)

In fact, one does not actually need stochastic dynamics in order for the dynamics of the distribution to be the meaningful object of study; random initial conditions can be enough. One of the best examples comes from perhaps the simplest and most famous example of a chaotic system: the logistic map. This example is described beautifully in [2].

Consider the following difference equation:

$$x[n+1] = 4x[n](1 - x[n]),$$

which we will study over the (invariant) interval $x \in [0, 1]$.

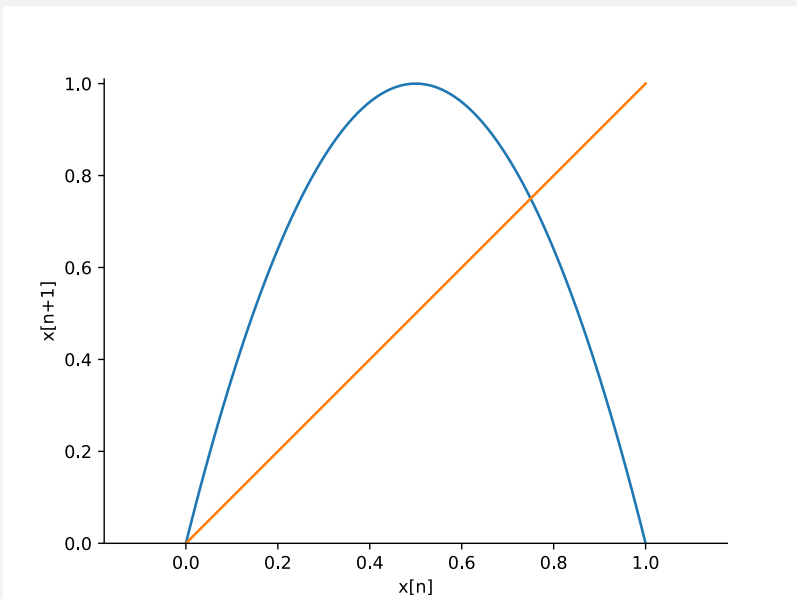

Figure 6.3 - The discrete-time dynamics of the "logistic map" (plotted alongside the line $x[n+1] = x[n]$).

It takes only a moment of tracing your finger along the plot using the "staircase method" to see what makes this system so interesting -- rollouts from a single initial condition end up bouncing all over the interval $(0, 1)$, and neighboring initial conditions will end up taking arbitrarily different trajectories (this is the hallmark of a "chaotic" system).

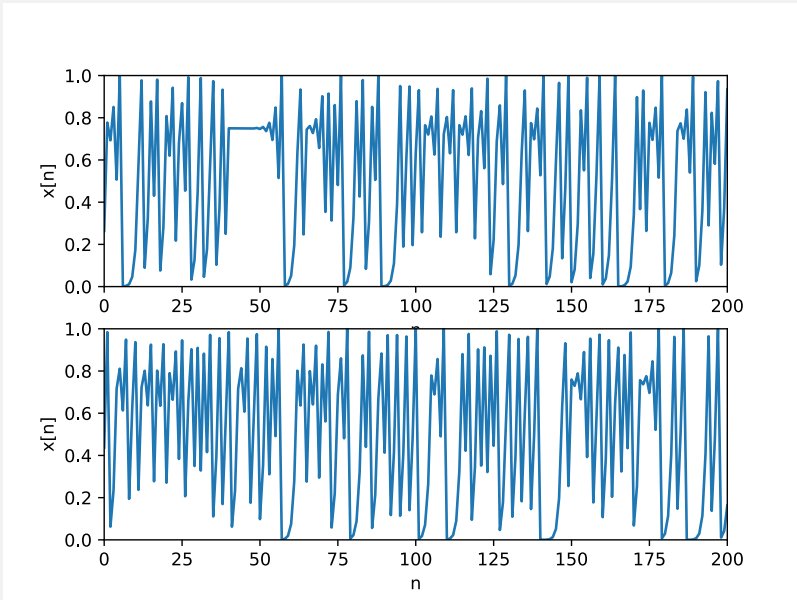

Figure 6.4 - Two simulations of the logistic map from different initial conditions. Remember, there is no stochasticity here -- the dynamics are entirely deterministic!

Here's what's completely fascinating -- even though the dynamics of any one initial condition for this system are extremely complex, if we study the dynamics of a distribution of states through the system, they are surprisingly simple and well-behaved. This system is one of the rare cases when we can write the master equation in closed form[2]:

$$p_{n+1}(x) = \frac{1}{4\sqrt{1-x}}\left[p_n\left(\frac{1}{2} - \frac{1}{2}\sqrt{1-x}\right) + p_n\left(\frac{1}{2} + \frac{1}{2}\sqrt{1-x}\right)\right].$$

Moreover, this master equation has a steady-state solution:

$$p_*(x) = \frac{1}{\pi\sqrt{x(1-x)}}.$$

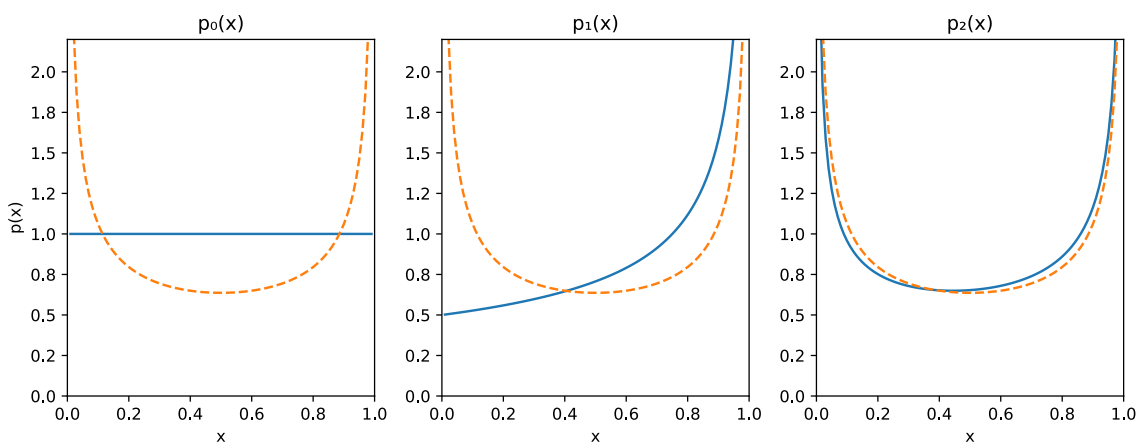

Figure 6.5 - Plotting the (closed-form) evolution of the master equation for the logistic map, initialized with $p_0(x) = 1$ for $x \in [0, 1]$, reveals surprising simplicity, and rapid convergence to a stationary distribution (dashed orange line).

For this system (and many chaotic systems), the dynamics of a single initial condition are complicated, but the dynamics a *distribution* of initial conditions are

beautiful and simple!

Note: when the dynamical system under study has deterministic dynamics (but a distribution of initial conditions), the linear map given by the master equation is known as the *Perron-Frobenius operator*, and it gives rise to the Liouville equation that we will study later in the text.

The slightly more general form of the master equation, which works for multivariate distributions with state-domain $\mathcal{X}$, and systems with control inputs $\mathbf{u}$, is

$$p_{n+1}(\mathbf{x}) = \int_{\mathcal{X}} p(\mathbf{x}|\mathbf{x}', \mathbf{u}) p_n(\mathbf{x}') d\mathbf{x}'.$$

This is a (continuous-state) Markov Decision Process.

Continuous-time formulations are also possible -- these lead to the so-called Fokker-Planck equation.

6.2 Stationary Distributions

In the example above, the histogram is our numerical approximation of the probability density. The logistic map example had the remarkable property that, although the individual trajectories of the system *do not* converge, the probability distribution actually *does* converge to what's known as a *stationary distribution* -- a fixed point of the master equation. Instead of thinking about the dynamics of the trajectories, we need to start thinking about the dynamics of the distribution.

The most important example of this analysis is for systems with linear dynamics and additive Gaussian noise; for this case we have closed-form solutions (in arbitrary dimensions). But to keep things simple, let's make sure we understand the scalar case first.

Example 6.3 (A One-dimensional Linear Gaussian System)

Let's consider the one-dimensional linear system with additive noise:

$$x[n+1] = ax[n] + w[n].$$

When $w[n] = 0$, the system is stable (to the origin) for $-1 < a < 1$. Let's make sure that we can understand the dynamics of the master equation for the case when $w[n]$ is Gaussian white noise with standard deviation σ.

First, recall that the probability density function of a Gaussian with mean μ is given by

$$p_w(w) = \frac{1}{\sqrt{2\pi\sigma_w^2}} e^{-\frac{(w-\mu)^2}{2\sigma_w^2}}.$$

When conditioned on $x[n]$, the distribution given by the dynamics subjected to mean-zero Gaussian white noise is simply another Gaussian, with the mean given by $ax[n]$:

$$p(x[n+1]|x[n]) = \frac{1}{\sqrt{2\pi\sigma_w^2}} e^{-\frac{(x[n+1]-ax[n])^2}{2\sigma_w^2}},$$

yielding the master equation

$$p_{n+1}(x) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \int_{-\infty}^{\infty} e^{-\frac{(x-ax')^2}{2\sigma_w^2}} p_n(x')dx'.$$

Now here's the magic. Let's push a distribution, $p_n(x)$, which is zero-mean, with standard deviation σ_n, through the master equation:

$$\begin{aligned} p_{n+1}(x) =& \frac{1}{\sqrt{2\pi\sigma_w^2}} \frac{1}{\sqrt{2\pi\sigma_n^2}} \int_{-\infty}^{\infty} e^{-\frac{(x-ax')^2}{2\sigma_w^2}} e^{-\frac{(x')^2}{2\sigma_n^2}} dx', \\ =& \frac{1}{\sqrt{2\pi(\sigma_w^2 + a^2\sigma_n^2)}} e^{-\frac{x^2}{2(\sigma_w^2+a^2\sigma_n^2)}}. \end{aligned}$$

The result is another mean-zero Gaussian with $\sigma_{n+1}^2 = \sigma_w^2 + a^2\sigma_n^2$. This result generalizes to the multi-variate case, too, and might be familiar to you e.g. from the process update of the Kalman filter.

Taking it a step further, we can see that a stationary distribution for this system is given by a mean-zero Gaussian with

$$\sigma_*^2 = \frac{\sigma_w^2}{1-a^2}.$$

Note that this distribution is well defined when $-1 < a < 1$ (only when the system is stable).

The stationary distribution of the linear Gaussian system reveals the fundamental and general balance between two terms in the governing equations of any stochastic dynamical system: the stability of the deterministic system is bringing trajectories together (smaller a means faster convergence of the deterministic system and results in a more narrow distribution), but the noise in the system is forcing trajectories apart (larger σ means larger noise and results in a wider distribution).

Given how rich the dynamics can be for deterministic nonlinear systems, you can probably imagine that the possible long-term dynamics of the probability are also extremely rich. If we simply flip the signs in the cubic polynomial dynamics we examined above, we'll get our next example:

Example 6.4 (The Cubic Example + Noise)

Now let's consider the discrete-time approximation of

$$\dot{x} = -x + x^3,$$

again with additive noise:

$$x[n+1] = x[n] + h(-x[n] + x[n]^3 + w[n]).$$

With $w[n] = 0$, the system has only a single stable fixed point (at the origin), whose deterministic region of attraction is given by $x \in (-1, 1)$. If we again simulate the system from a set of random initial conditions and plot the histogram, we will see something like the figure below.

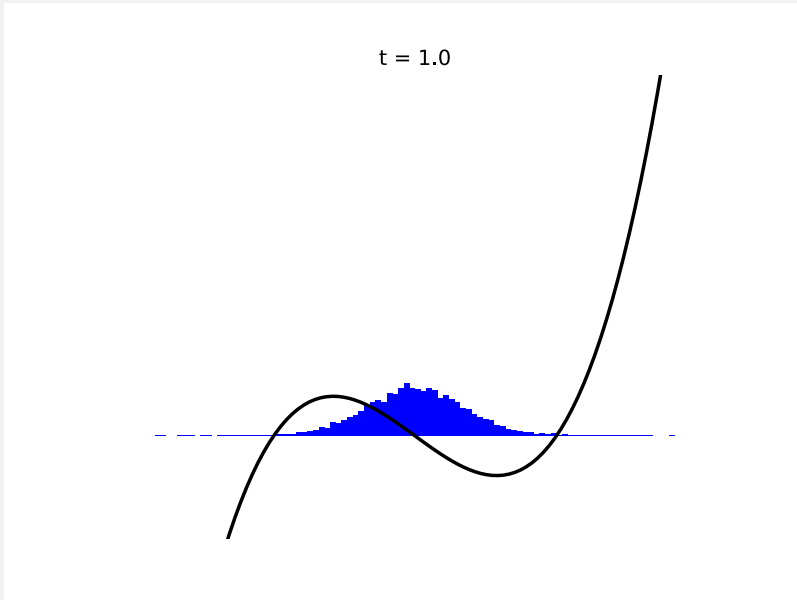

Figure 6.6 - Histogram of the states of the bistable system with noise ($h = 0.1, \sigma = 1.5$) after simulating from random initial conditions until $t = 1$.
Figure 6.7 - Histogram of the states of the bistable system with noise ($\sigma = 1.5$).

Be sure to watch the animation. Or better yet, run the simulation for yourself by changing the sign of the derivative in the bistability example and re-running:

You can run this demo for yourself:

What is the stationary distribution for this system? In fact, there isn't one. Although we initially collect probability density around the stable fixed point, you should notice a slow leak -- on every step there is some probability of transitioning past unstable fixed points and getting driven by the unstable dynamics off towards infinity. If we run the simulation long enough, there won't be any probability density left at $x = 0$.

Example 6.5 (The Stochastic Van der Pol Oscillator)

One more example; this is a fun one. Let's think about one of the simplest examples that we had for a system that demonstrates limit cycle stability -- the Van der Pol oscillator -- but now we'll add Gaussian white noise,

$$\ddot{q} + (q^2 - 1)\dot{q} + q = w(t),$$

Here's the question: if we start with a small set of initial conditions centered around one point on the limit cycle, then what is the long-term behavior of this distribution?

Figure 6.8 - Randomly sampled initial conditions pictured with the stable limit cycle of the Van der Pol oscillator.

Since the long-term behavior of the deterministic system is periodic, it would be very logical to think that the state distribution for this stochastic system would fall into a stable periodic solution, too. But think about it a little more, and then watch the animation (or run the simulation yourself).

Figure 6.9 - Rollouts of the stochastic Van der Pol oscillator.

The explanation is simple: the periodic solution of the system is only *orbitally stable*; there is no stability along the limit cycle. So any disturbances that push the particles along the limit cycle will go unchecked. Eventually, the distribution will "mix" along the entire cycle. Perhaps surprisingly, this system that has a limit cycle stability when $w = 0$ eventually reaches a stationary distribution (fixed point) in the master equation.

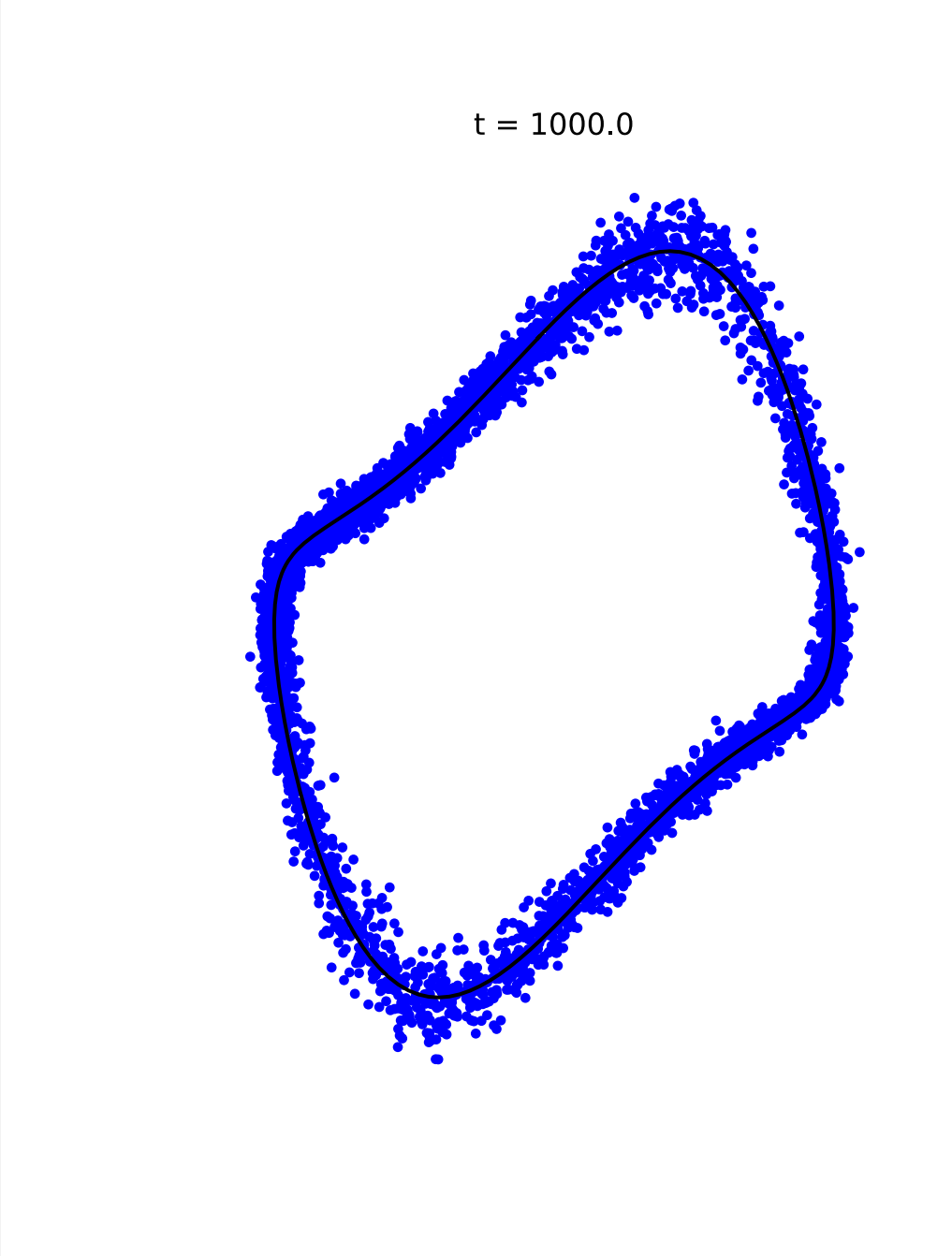

Figure 6.10 - Sampled approximation of the stationary distribution of the stochastic Van der Pol oscillator.

6.3 Extended Example: The Rimless Wheel on Rough Terrain

My favorite example of a meaningful source of randomness on a model underactuated system is the rimless wheel rolling down stochastically "rough" terrain[1]. Generating interesting/relevant probabilistic models of terrain in general can be quite complex, but the rimless wheel makes it easy -- since the robot only contacts that ground at the point foot, we can model almost arbitrary rough terrain by simply taking the ramp angle, γ, to be a random variable. If we restrict our analysis to rolling only in one direction (e.g. only downhill), then we can even consider this ramp angle to be i.i.d.; after each footstep we will independently draw a new ramp angle $\gamma[n]$ for the next step.

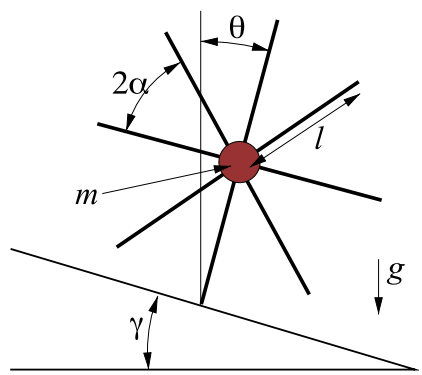

Figure 6.11 - The rimless wheel. The orientation of the stance leg, θ, is measured clockwise from the vertical axis.

In our original analysis of the rimless wheel, we derived the "post-collision" return map -- mapping the angular velocity from the beginning of the stance phase to the (post-collision) angular velocity at the next stance phase. But now that the relative location of the ground is changing on every step, we instead write the "apex-to-apex" return map, which maps the angular velocity from one apex (the moment that the leg is vertical) to the next, which is given by:

$$\dot{\theta}[n+1] = \sqrt{\cos^2\alpha\left(\dot{\theta}^2[n] + \frac{2g}{l}(1-\cos(\alpha+\gamma[n]))\right) - \frac{2g}{l}(1-\cos(\alpha-\gamma[n]))}.$$

More coming soon. Read the paper [1] and/or watch the video.

6.4 Noise models for real robots/systems.

Sensor models. Beam model from probabilistic robotics. RGB-D dropouts.

Perception subsystem. Output of a perception system is not Gaussian noise, it's missed detections/drop-outs...

Distributions over tasks/environments.

References

1. Katie Byl and Russ Tedrake, "Metastable Walking Machines", *International Journal of Robotics Research*, vol. 28, no. 8, pp. 1040-1064, August 1, 2009. [link]

2. Andrzej Lasota and Michael C Mackey, "Chaos, fractals, and noise: stochastic aspects of dynamics", Springer Science \& Business Media , vol. 97, 2013.

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